AQA GCSE Biology

BIOENERGETICS

Complete Revision Notes

Foundation & Higher Tier | [H] = Higher Only

1. Photosynthesis

1.1 What is Photosynthesis?

Photosynthesis is the process by which plants, algae and some bacteria convert light energy into chemical energy stored in glucose. It is an endothermic reaction — it takes in energy from the surroundings.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ carbon dioxide + water → glucose + oxygen
(Light energy and chlorophyll required)

Reactants	Carbon dioxide (CO_2) absorbed from air through stomata; Water (H_2O) absorbed from soil through roots

Products	Glucose ($C_6H_{12}O_6$) used for energy and building materials; Oxygen (O_2) released as a by-product

Energy source	Light energy (from the Sun) absorbed by chlorophyll in chloroplasts

⚠EXAMINER WARNING: Photosynthesis is ENDOTHERMIC — it takes IN energy. Do not confuse with respiration which is exothermic (releases energy).

1.2 The Chloroplast

Photosynthesis takes place in the chloroplasts. Chloroplasts contain a green pigment called chlorophyll which absorbs light energy.

- Chlorophyll absorbs red and blue light most effectively
- Green light is reflected — this is why plants appear green
- Chloroplasts are found mainly in leaf mesophyll cells
- [H] The light-dependent stage occurs at the thylakoid membranes
- [H] The light-independent stage (Calvin cycle) occurs in the stroma

1.3 Uses of Glucose in Plants

Use of Glucose	Details
Respiration	Glucose is respired to release energy for all cell processes
Making cellulose	Used to make cell walls for growth and support
Making amino acids	Combined with nitrate ions from soil to make amino acids for proteins

Making starch	Stored as insoluble starch in roots, stems and leaves
Making oils/fats	Stored in seeds for energy
Making sucrose	Transported around the plant in the phloem

⚠EXAMINER WARNING: Starch is the STORAGE form of glucose in plants. It is insoluble so does not affect osmosis. Glucose itself is converted to starch for storage.

1.4 Limiting Factors of Photosynthesis

A limiting factor is something that, when in short supply, prevents the rate of photosynthesis from increasing further even if other conditions are ideal.

Limiting Factor	Effect when increased	Effect when too low
Light intensity	Rate of photosynthesis increases (up to a point)	Rate decreases — not enough energy to drive reactions
CO_2 concentration	Rate increases (up to a point)	Rate slows — shortage of raw material
Temperature	Rate increases (enzymes work faster)	Rate decreases; above optimum enzymes denature
[H] Chlorophyll	More chlorophyll = more light absorbed = faster rate	Less chlorophyll limits light absorption

⚠EXAMINER WARNING: In a graph showing rate vs light intensity: if the curve levels off (plateau), another factor (CO_2 or temperature) is now the limiting factor — NOT light.

1.5 [H] Inverse Square Law

[H] Light intensity is inversely proportional to the square of the distance from the light source:

light intensity □ $1/d^2$ where d = distance from light source

[H] This means if you double the distance, the light intensity falls to one quarter. Used in the photosynthesis required practical.

2. Respiration

2.1 What is Respiration?

Respiration is the process by which organisms break down glucose to release energy for use in cell processes. It occurs continuously in ALL living cells. It is an EXOTHERMIC reaction — it releases energy to the surroundings.

Energy uses	Muscle contraction (movement); keeping warm (maintaining body temperature); active transport; building molecules; growth

2.2 Aerobic Respiration

Aerobic respiration requires oxygen. It releases the maximum amount of energy from glucose.

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ (+ energy released) glucose + oxygen → carbon dioxide + water

- Occurs in the mitochondria (main site)
- Releases much more energy than anaerobic respiration
- Products: carbon dioxide and water
- Requires continuous oxygen supply

2.3 Anaerobic Respiration

Anaerobic respiration occurs WITHOUT oxygen. It releases less energy than aerobic respiration but is faster and allows activity to continue when oxygen is limited.

In animals (including humans):

glucose → lactic acid (+ small amount of energy) Occurs in muscle cells during intense exercise

In plants and yeast (fermentation):

glucose → ethanol + carbon dioxide (+ small amount of energy) This is used in brewing and bread-making

Feature	Aerobic Respiration	Anaerobic Respiration
Oxygen needed?	Yes	No
Products (animals)	$CO_2 + H_2O$	Lactic acid
Products (plants/yeast)	$CO_2 + H_2O$	Ethanol + CO_2
Energy released	Much more (maximum)	Much less
Location in cell	Mitochondria	Cytoplasm
When used	Normal conditions	During intense exercise / limited oxygen

⚠EXAMINER WARNING: Anaerobic respiration in ANIMALS produces LACTIC ACID (not ethanol). Anaerobic respiration in YEAST/PLANTS produces ETHANOL + CO_2. These are different — a very common exam error.

2.4 Metabolism

Metabolism is the sum of ALL chemical reactions in a cell or organism. It includes:

- Aerobic and anaerobic respiration
- Synthesis of large molecules from smaller ones (e.g. proteins from amino acids)
- Breaking down excess amino acids (deamination in the liver)
- Reactions of photosynthesis in plants

2.5 [H] Response to Exercise

[H] During exercise, muscles require more energy. The body responds by:

- [H] Increasing heart rate — to deliver more oxygen and glucose to muscles
- [H] Increasing breathing rate and depth — to take in more oxygen and remove CO_2
- [H] Increasing blood flow to muscles (vasodilation of arterioles)

[H] If the demand for oxygen exceeds supply, muscles respire anaerobically, producing lactic acid.

2.6 [H] Oxygen Debt

[H] After intense exercise, lactic acid has built up in muscles. Extra oxygen is needed to break down lactic acid. This extra oxygen requirement is called the oxygen debt (or excess post-exercise oxygen consumption, EPOC).

> [H] Lactic acid is transported in the blood to the liver where it is converted back to glucose using oxygen. This is why we continue to breathe rapidly after exercise ends.

> **⚠EXAMINER WARNING: [H] Lactic acid is NOT 'burned off' — it is converted back to glucose by the liver using oxygen. The oxygen debt is the amount of oxygen needed to do this.**

3. Required Practicals

3.1 Required Practical: Rate of Photosynthesis

Aim:

To investigate the effect of light intensity on the rate of photosynthesis in aquatic plants (e.g. Elodea or Cabomba).

Method (counting bubbles):

1. Place a piece of Elodea in a beaker of sodium hydrogen carbonate solution (provides CO_2)
2. Position a lamp at a measured distance from the plant
3. Count the number of oxygen bubbles produced per minute
4. Repeat at different distances from the light source
5. Use a ruler to measure distance; keep temperature and CO_2 concentration constant

Variable	Type	How Controlled/Measured
Distance from light source	Independent	Changed using a ruler (e.g. 10, 20, 30, 40 cm)
Rate of photosynthesis (bubbles/min)	Dependent	Count oxygen bubbles per minute or use gas syringe
Temperature	Control	Use water bath or monitor with thermometer
CO_2 concentration	Control	Keep sodium hydrogen carbonate concentration constant
Type/size of plant	Control	Same species and similar size of plant used throughout

Results expected:

- As distance increases, light intensity decreases (inverse square law [H])
- As light intensity decreases, rate of photosynthesis (bubble count) decreases
- [H] Rate is proportional to $1/d^2$

⚠EXAMINER WARNING: Counting bubbles is imprecise — bubble size may vary. More accurate: use a gas syringe to measure the volume of oxygen produced.

3.2 Required Practical: Rate of Respiration

Aim:

To investigate the effect of temperature on the rate of respiration using a respirometer.

Method (using a respirometer with woodlice or seeds):

6. Set up respirometer with living organisms (e.g. germinating seeds) in one side and dead seeds on the other as a control
7. Add coloured fluid/dye to a capillary tube connected to each chamber
8. As organisms respire, they consume O_2 — CO_2 is absorbed by soda lime
9. The decrease in gas volume causes the dye to move towards the organism
10. Measure the distance the dye moves in a set time
11. Repeat at different temperatures using a water bath

Variable	Type	How Controlled/Measured
Temperature	Independent	Changed using a water bath; measured with thermometer
Rate of O_2 consumption (dye movement)	Dependent	Measure distance dye moves per minute
Number/mass of organisms	Control	Same mass of organisms in each trial
Volume of chamber	Control	Same respirometer equipment used throughout
CO_2 removal	Control	Soda lime absorbs CO_2 so only O_2 change is measured

Results expected:

- As temperature increases (up to enzyme optimum ~37-40°C), rate of respiration increases
- Above optimum temperature, enzymes begin to denature and rate decreases
- Dead seeds show no movement of dye (negative control)

⚠EXAMINER WARNING: Soda lime is hazardous — handle with care and wear eye protection. Ensure the system is airtight otherwise dye movement will not accurately reflect oxygen uptake.

4. Key Terms Glossary

Term	Definition
Photosynthesis	Process by which plants use light energy to convert CO_2 and H_2O into glucose and O_2
Endothermic reaction	A reaction that takes in energy from the surroundings
Chlorophyll	Green pigment in chloroplasts that absorbs light energy for photosynthesis
Limiting factor	A factor that limits the rate of a reaction when it is in short supply
Respiration	Process by which organisms break down glucose to release energy
Exothermic reaction	A reaction that releases energy to the surroundings
Aerobic respiration	Respiration using oxygen — releases maximum energy; products: CO_2 + H_2O
Anaerobic respiration	Respiration without oxygen — releases less energy; products: lactic acid (animals) or ethanol + CO_2 (plants/yeast)
Fermentation	Anaerobic respiration in yeast/plants producing ethanol and CO_2
Metabolism	The sum of all chemical reactions in a cell or organism
[H] Oxygen debt	Extra oxygen needed after exercise to convert lactic acid back to glucose
[H] Lactic acid	Waste product of anaerobic respiration in animals; causes muscle fatigue and cramp
Starch	Insoluble storage form of carbohydrate in plants
Stomata	Tiny pores in leaves through which CO_2 enters and O_2 and water vapour exit
Chloroplast	Organelle in plant cells where photosynthesis occurs; contains chlorophyll
[H] Inverse square law	Light intensity $\propto 1/d^2$; doubling distance reduces intensity to one quarter